Semiconductor RAM Memories: A static RAM cell

Random Access Memory

RAM is a **read/write memory** where the data can be read from or written into any of the memory locations regardless of the order in which they are arranged.

Depending upon the nature of the memory cell used, there are **two types of RAM**, ***namely Static RAM (SRAM) and Dynamic RAM (DRAM).***

In **SRAM**, the memory cell is essentially a latch and can store data indefinitely as long as the DC power is supplied.

DRAM on the other hand, has a memory cell that stores data in the form of charge on a capacitor. Therefore, DRAM cannot retain data for long and hence needs to be refreshed periodically. SRAM has a higher speed of operation than DRAM but has a smaller storage capacity.

Both are Volatile Memory because Data lost when power is off.

Memories that consist of circuits capable of retaining their state as long as power is applied are known as ***static memories.*** **Two cross-coupled inverters** Form a **bistable latch**

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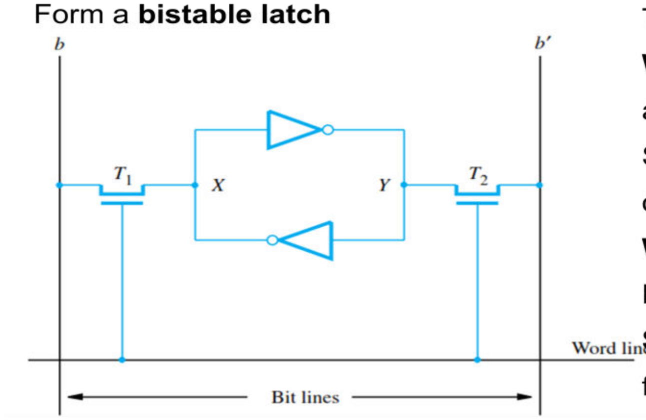


1-bit static RAM cell

T1, T2 : Transistors

Word line =0, the transistors are turned off and the latch retains its state.

State is maintained as long as the signal on the word line is at ground level.

Word line=1, ON T1 and T2 to perform Read or Write operation.

Sense/ Write circuit: decide to read data from data line (b/b') or data to be write through data line

Working principle: Hold state

➢ Hold State (Memory Retention)

Word line = 0 → T1, T2 OFF
Cell is isolated from bit lines
Cross-coupled inverters maintain value:
If **X = 1 → Y = 0**
If **X = 0 → Y = 1**

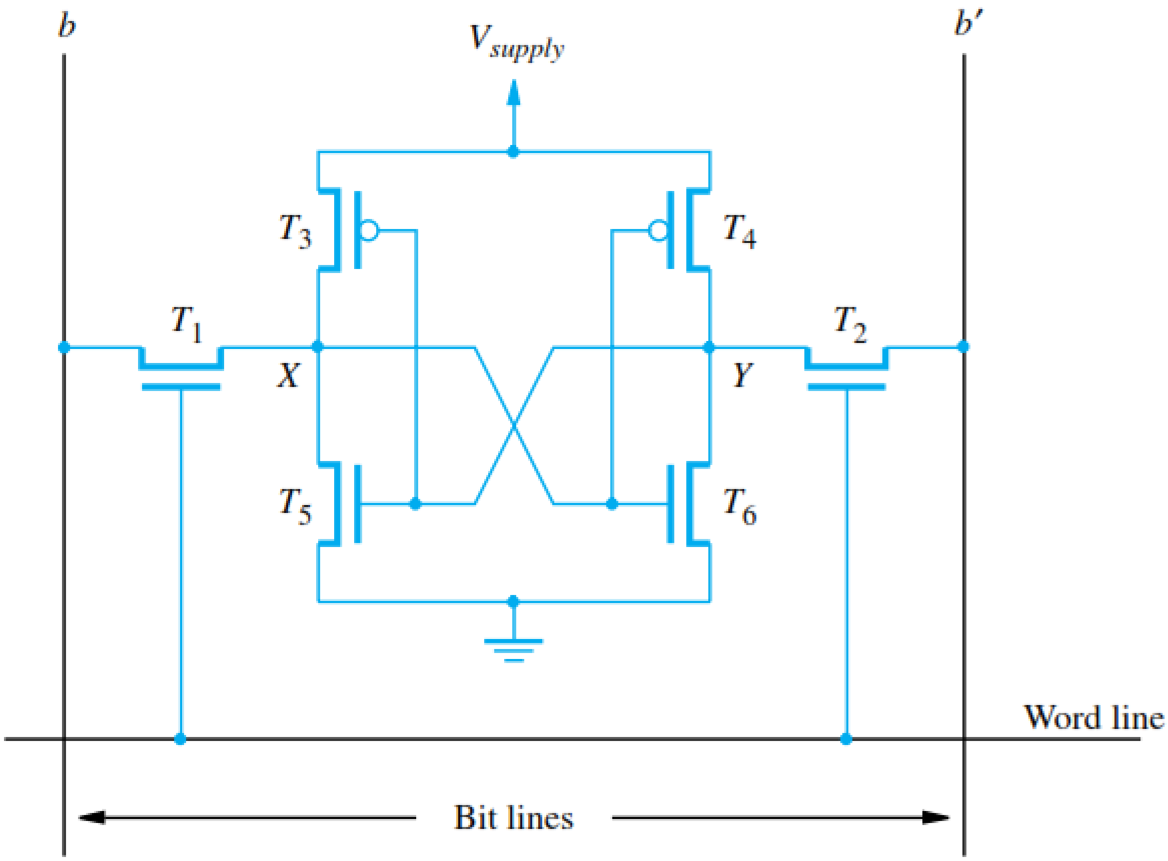
This is why SRAM is called **static** memory

Working principle: Read Operation

- Word line = 1 → T1, T2 ON
- If the cell is in state 1, the signal on bit line b is high and the signal on bit line b’ is low. The opposite is true if the cell is in state 0.
- Thus, b and b’ are always complements of each other.
- The Sense/Write circuit at the end of the two bit lines monitors their state and sets the corresponding output accordingly.

Working principle: Write Operation

- Word line = 1 → T1, T2 ON
- Apply data on bit lines
- It places the appropriate value on bit line b and its complement on b’ and activates the word line.
- This forces the cell into the corresponding state, which the cell retains when the word line is deactivated.



- ✓ SRAMs are said to be ***volatile*** memories because their contents are lost when power is interrupted.
- ✓ **A major advantage of CMOS SRAMs** is their very low power consumption, because current flows in the cell only when the cell is being accessed.
- ✓ Static RAMs can be accessed very quickly. Access times on the order of a few nano seconds are found in commercially available chips.
- ✓ Each cells require several transistors.

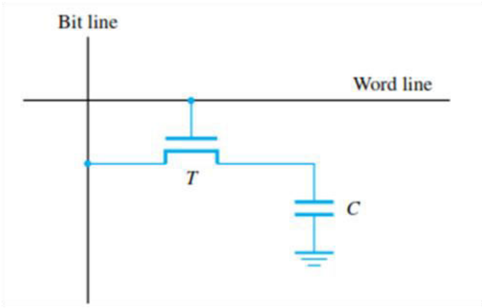
Dynamic RAMs

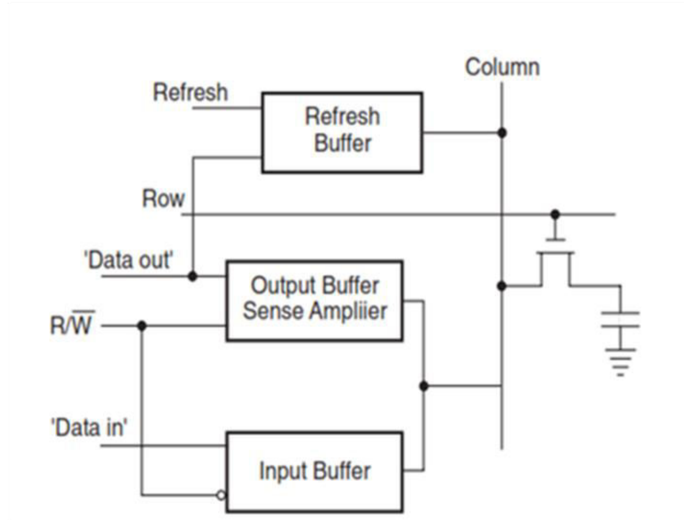
- Less expensive and higher density RAMs can be implemented with simpler cells. But, these simpler cells do not retain their state for a long period, unless they are accessed frequently for Read or Write operations.
- Memories that use such cells are called *dynamic RAMs (DRAMs).*
- **DRAM : SLOW,CHEAP AND DENSE MEMORY.**
- **Typical choice of main memory.**

Dynamic RAMs

- Information is stored in a dynamic memory cell in the form of a charge on a capacitor, but this charge can be maintained for only **tens of milliseconds.**
- Since the cell is required to store information for a much longer time, its contents must be **periodically refreshed** by restoring the capacitor charge to its full value.
- This occurs when the contents of the cell are read or when new information is written into it.

Dynamic RAMs





A single-transistor dynamic memory cell.

Dynamic RAMs

Why Periodic refresh is required?

After the transistor is turned off, the capacitor begins to discharge.

1. This happens due to the capacitors own leakage resistance.

2. The transistor continues to conduct a tiny amount of current measured in Pico amperes, after it is turned off.

Dynamic RAMs: How Periodic Refreshment operation is performed?

- ❖ During a Read operation, the transistor in a selected cell is turned on.
- ❖ A sense amplifier connected to the bit line detects whether the charge stored in the capacitor is above or below the threshold value.
- ❖ If the charge is above the threshold, the sense amplifier drives the bit line to the full voltage representing the logic value 1.
- ❖ As a result, the capacitor is recharged to the full charge corresponding to the logic value 1.
- ❖ If the sense amplifier detects that the charge in the capacitor is below the threshold value, it pulls the bit line to ground level to discharge the capacitor fully, representing the logic value 0.
- ❖ Thus, reading the contents of a cell automatically refreshes its contents.
- ❖ Since the word line is common to all cells in a row, all cells in a selected row are read and refreshed at the same time.

Difference between S-RAM and D-RAM

S-RAM	D-RAM
Binary cell made up of Latch.	Binary Cell is made-up of capacitors and transistors
In SRAM, it can hold data indefinitely until power is on in the circuit.	It can not hold data indefinitely even if power is kept on to the Circuit. (Due to charging a discharging of capacitor)
No need of periodical refreshment.	Need of periodical refreshment.
Costly and complex to design.	simple to design and very low cost.
It can store large no. of data using of Binary Cell.	The no. of data that can be stored is higher compared to using S-RAM.
Static RAMs are fast, hence its applications where speed is of critical concern.	It is Less expensive and higher density hence its applications where large storage capacity is required.

Interleaved memory

Interleaved memory is a memory organization technique where main memory is divided into multiple **banks (modules)**, allowing **parallel access** to increase memory bandwidth and reduce access time.

Instead of storing consecutive addresses in the same memory module, interleaving distributes them across different banks.

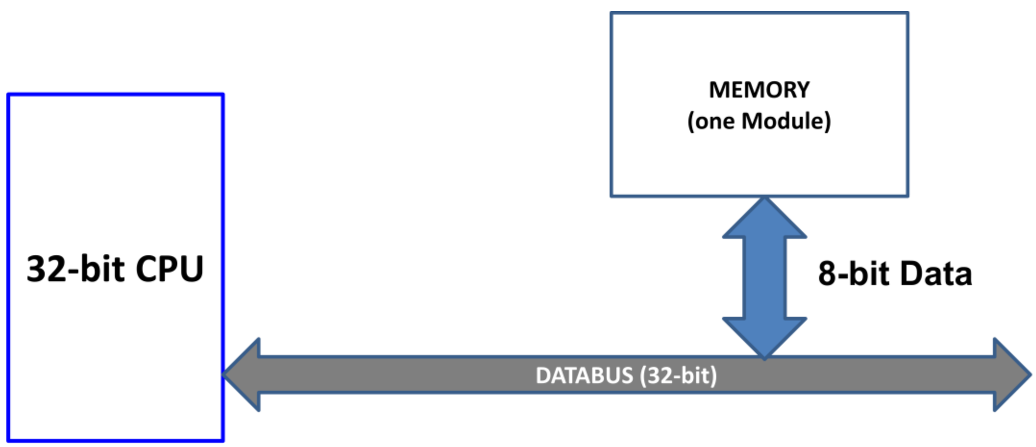
One way to speed up access to main memory.

A two-way interleave means that main memory has been divided into two sections.

A four-way interleave means that main memory has been divided into four sections.

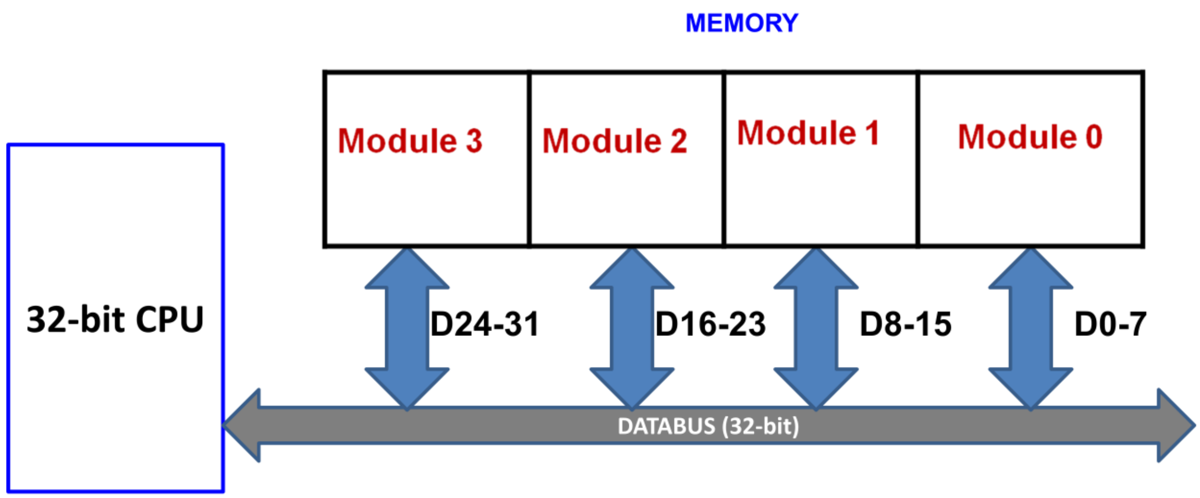
It is a technique which divides memory into a number of modules such that successive words in the address space are placed in the different module. Each module is called Memory bank.

- Main memory divided into **two or more sections.**
- The CPU can access **alternate sections immediately**, without **waiting for memory** to catch up (through wait states).
- **Interleaved memory is one technique for compensating for the relatively slow speed of dynamic RAM (DRAM).**
- Memory interleaving **increases bandwidth** by allowing simultaneous access to more than one chunk of memory.
- This improves performance because the processor can transfer more information to/from memory in the same amount of time.



CPU word length = 32-bit, Memory Module word length = 8-bit

To read/write 1 word from memory _____4_____ no. of memory access required because one word in one module.



CPU word length = 32-bit, Memory word length = 8-bit

To read/write 1 word from memory ____1____ no. of memory access required because one word in 4 module.

Types of Memory Interleaving

1. Low order interleaving:

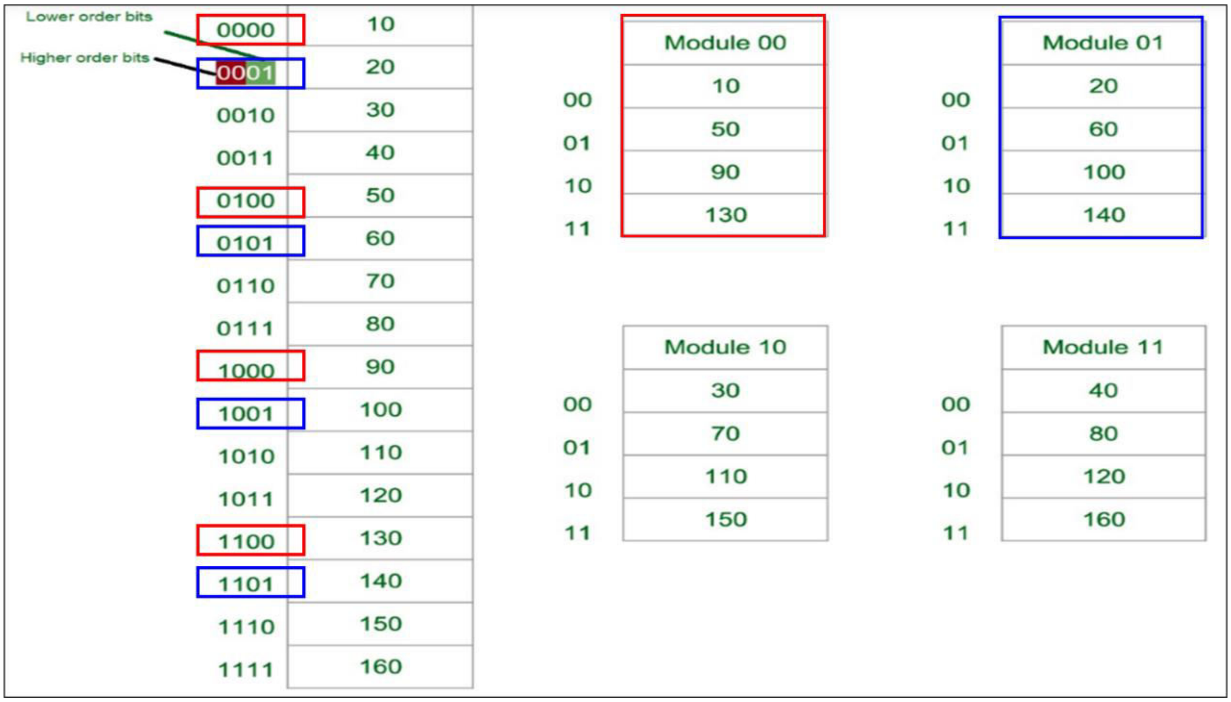
The least significant bits select the memory bank (module) in low-order interleaving.

Address bits	Upper address Bits	Lower address Bits
Use	Address to the chip	Bank Select

Module 0	Module 1	Module 2	Module 3
0	1	2	3
8	9	10	11
16	17	18	19
24	25	26	27

In this, consecutive memory addresses are in different memory modules, allowing memory access faster than the cycle time.

Low order interleaving : Consecutive Word in Consecutive Module



Now again we assume 16 Data's to be transferred to the **Four Module**. But **Now the consecutive Data are added in different Consecutive Module.** That is, 10 [Data] is added in Module 1, 20 [Data] in Module 2 and So on.

Least Significant Bit (LSB) provides the Address of the Module & Most significant bit (MSB) provides the address of the data in the module.

Module 1 Contains Data : 10, 50, 90, 130
Module 2 Contains Data : 20, 60, 100, 140
Module 3 Contains Data : 30, 70, 110, 150
Module 4 Contains Data : 40, 80, 120, 160

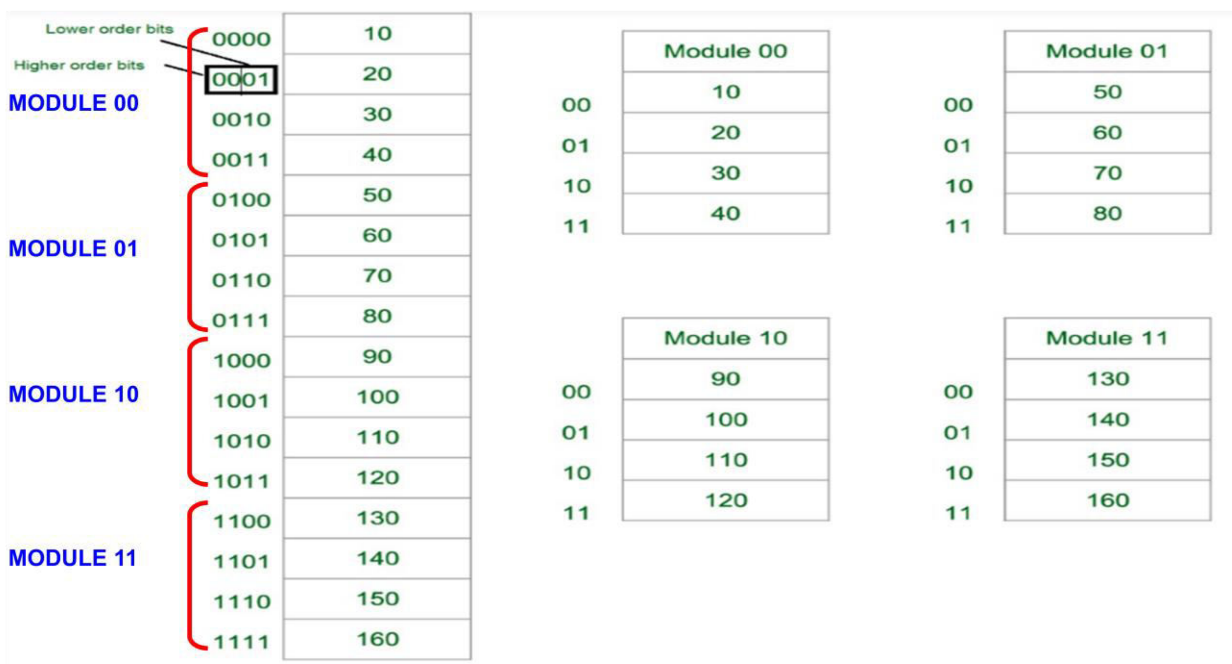
1. High order interleaving:

- In high order memory interleaving, the most significant bits of the memory address decides memory banks where a particular location resides.

- The least significant bits are sent as addresses to each chip.

One problem is that consecutive addresses tend to be in the same chip. The maximum rate of data transfer is limited by the memory cycle time.

High order interleaving : Consecutive Word in a Module



From the figure above in Module 1, 10 [Data] is transferred then 20, 30 & finally, 40 which are the Data. **That means the data are added consecutively in the same Module till its max capacity**. Most significant bit (MSB) provides the Address of the Module & least significant bit (LSB) provides the address of the data in the module.

For **Example**, to get 90 (Data) 1000 will be provided by the processor. In this 10 will indicate that the data is in module 10 (module 3) & 00 is the address of 90 in Module 10 (module 3). So,

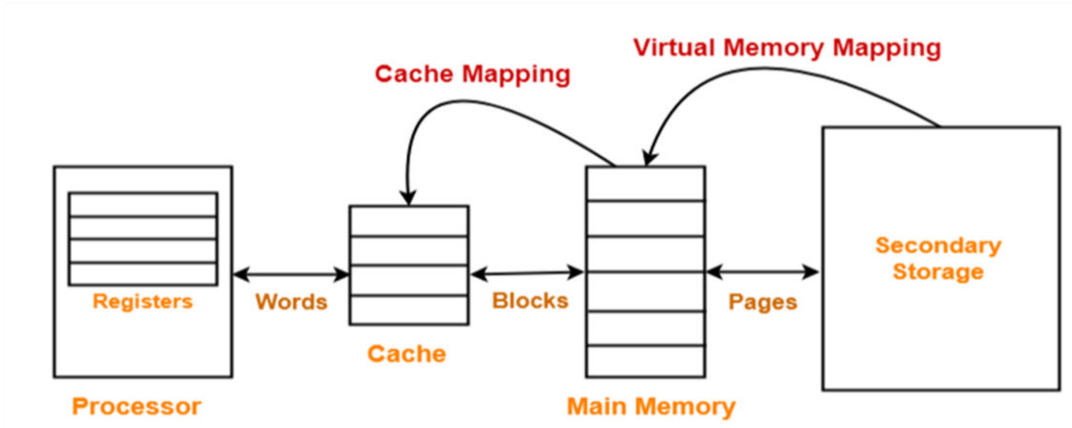
Module 1 Contains Data : 10, 20, 30, 40
Module 2 Contains Data : 50, 60, 70, 80
Module 3 Contains Data : 90, 100, 110, 120
Module 4 Contains Data : 130, 140, 150, 160

Why we use Memory Interleaving?

Whenever, **Processor request Data from** the **main memory**, **A block** (chunk) of data is transferred to the **cache and then to processor**. So whenever a cache miss occurs the data is to be fetched from main memory. But main memory is relatively slower than the cache. So to improve the access time of the main memory interleaving is used.

We can access all four module at the same time thus **achieving parallelism**.

Different Memories – Mapping Basics



Cache memory bridges the speed mismatch between the processor and the main memory.

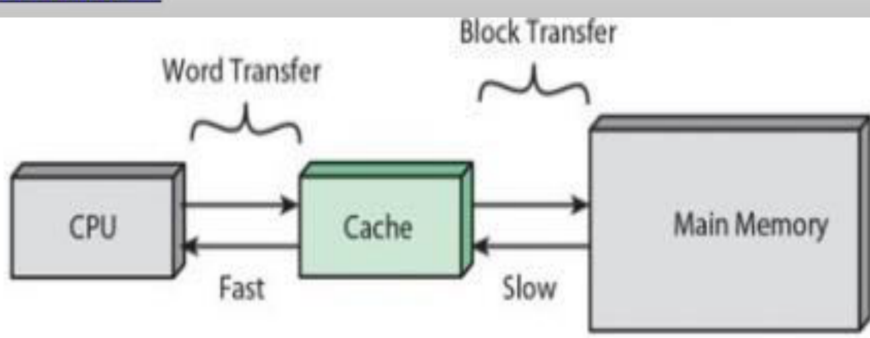
Why Cache Memory is required ?

Processor is much faster than the main memory.

- As a result, the processor has to spend much of its time waiting while instructions and data are being fetched from the main memory.
- Major obstacle towards achieving good performance.

Speed of the main memory cannot be increased beyond a certain point.

Cache memory is based on the property of computer programs known as **"locality of reference".**



Locality of reference is the tendency of a CPU to access the **same memory locations** or **nearby locations** repeatedly within a short time. This principle is the **foundation of cache memory design**

Types of Locality

Temporal Locality (Time-Based)

If a memory location is accessed now, it is likely to be accessed again soon

Example:

```
For (int i=0;i<1000;i++) {
   sum = sum + 5;   // 'sum' reused repeatedly
}
```

- ❖ Same variable accessed multiple times
  Stored in cache → faster execution

Spatial Locality (Space-Based)

If a memory location is accessed, nearby addresses will likely be accessed soon

Example:

```
For (int i=0;i<1000;i++) {
   sum =sum+ arr[i];   // sequential array access
}
```

arr[i], arr[i+1], arr[i+2] are contiguous
Entire block fetched into cache

Temporal Locality of reference : It is based on time

- ❖ A recently executed instruction is likely to be executed again very soon.
- ❖ If an instruction is executed at time instant t, then most likely the same instruction will be executed at time instant $t + \Delta t$.

Spatial Locality of reference

- ❖ The spatial aspect means that instructions close to a recently instruction are also likely to be executed soon.
- ❖ If an instruction 'I' is executed at time instant t, then most likely the instruction at address 'i+1' will be executed at time instant $t + \Delta t$.

What is Cache Memory?

- Small and fast (SRAM) memory technology
- Stores the subset of instructions & data currently being accessed.
- Used to reduce average access time to memory.
- Caches exploit **temporal locality** by Keeping recently accessed data closer to the processor.
- Caches exploit **spatial locality** by Moving blocks consisting of multiple contiguous words.
- Cache memory sits between CPU and main memory

Cache memory working

- Cache memory is logically divided into blocks or lines, where every block (line) typically contains no. of bytes.
- When the CPU wants to access a word in memory, a special hardware first checks whether it is present in cache memory.
  - If yes (called **cache hit**), the word is directly accessed from the cache memory.
  - If not (called **cache miss**), the block containing the requested word is brought from main memory to cache.
  - For writes, sometimes the CPU can also directly write to main memory.
- Objective is to keep the commonly used blocks in the cache memory.
  - Will result in significantly improved performance due to the property of **locality of reference.**

- At any given time, only some blocks in the main memory are held in the cache.
- **Mapping functions** determine where a main memory block will be placed in the cache.
- When the cache is full, and a block of words needs to be transferred from the main memory, some block of words in the cache must be replaced. This is determined by a **"replacement algorithm".**

Levels of Cache Memory

L1: Level 1 cache

- First level of cache
- Memory is present inside the CPU itself.
- It can work at the same speed as of the CPU.
- Each core of CPU have its own level 1 cache.

L2: Level 2 cache

- Second level of cache
- It can be present inside or outside the CPU.
- They are slower than L1.
- Each core of CPU can have its own L2 cache or they can share single L2 cache.

L3: Level 3 cache

- Third level of cache
- It is located outside the CPU.
- It is slower than L1 & L2.
- It is shared by all the cores of a CPU.

Cache mapping

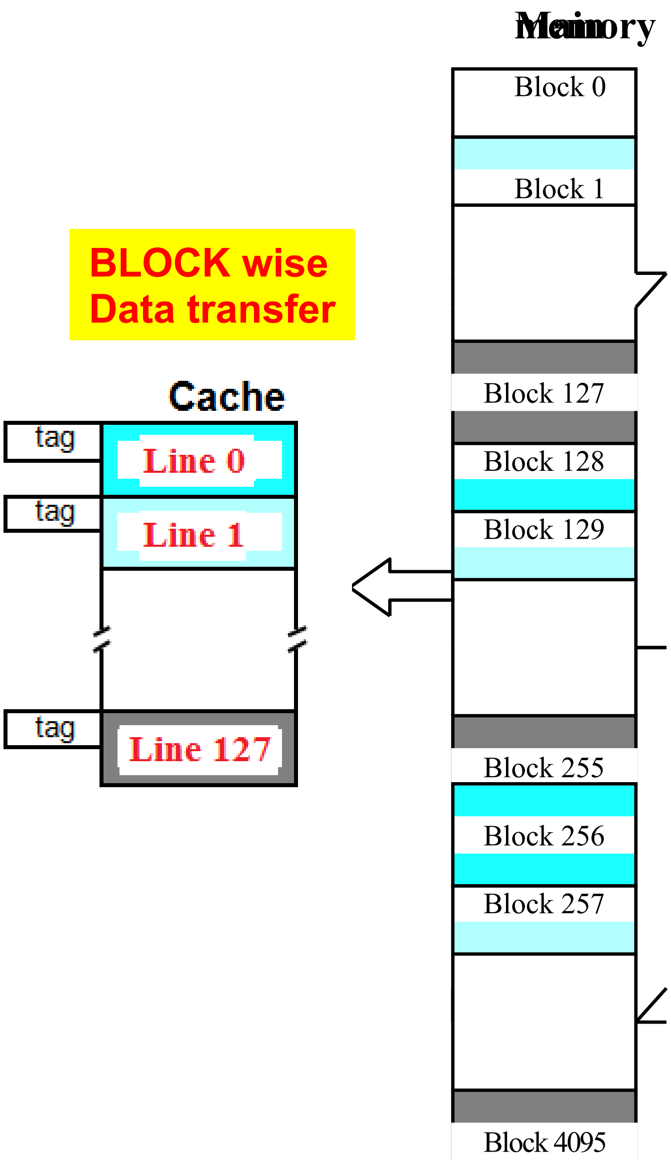
It is also called Block placement.

What is a Block?

A **block** is a set of consecutive memory locations content.

- ❖ When processor generates an address for any read/write operation, first it check whether that address's content is present in the cache memory or not.
- ❖ If yes, then we perform our read/write operation from the cache memory.
- ❖ If not, then we bring the block containing the address that we are trying to access for our read/write operation from the main memory. i.e., the unit of transfer between main memory and cache is **a block.**
- ❖ The main memory is bigger compared to the cache, hence the length of the main memory address is more compared to the cache memory's address.
- ❖ In block placement/ mapping functions we will see how to access a byte within a block in the cache memory, for a given main memory address.

Cache mapping



- Main memory divided into No. of Blocks.
- Each Block consists of No. of words or Bytes.
- 1 Word/Byte refer to one address
- Cache memory divided into no. of blocks called Lines.
- Block size = Line size
- Main memory block will be mapped in cache line
- Each cache line having one tag no. along with groups of words or bytes.

Cache Line	
Tag No.	Words (Data)

1. Draw and discuss how read and write operation carried out in 1-bit SRAM cell. What are the different characteristics of it?
2. Differentiate between SRAM and DRAM. Why is SRAM used in cache and DRAM in main memory?
3. Explain low-order and high-order interleaving with example.
4. Define cache memory. Why is it required in computer systems? Explain its working principle using locality of reference, cache hit, and miss. Also describe different levels of cache memory.